

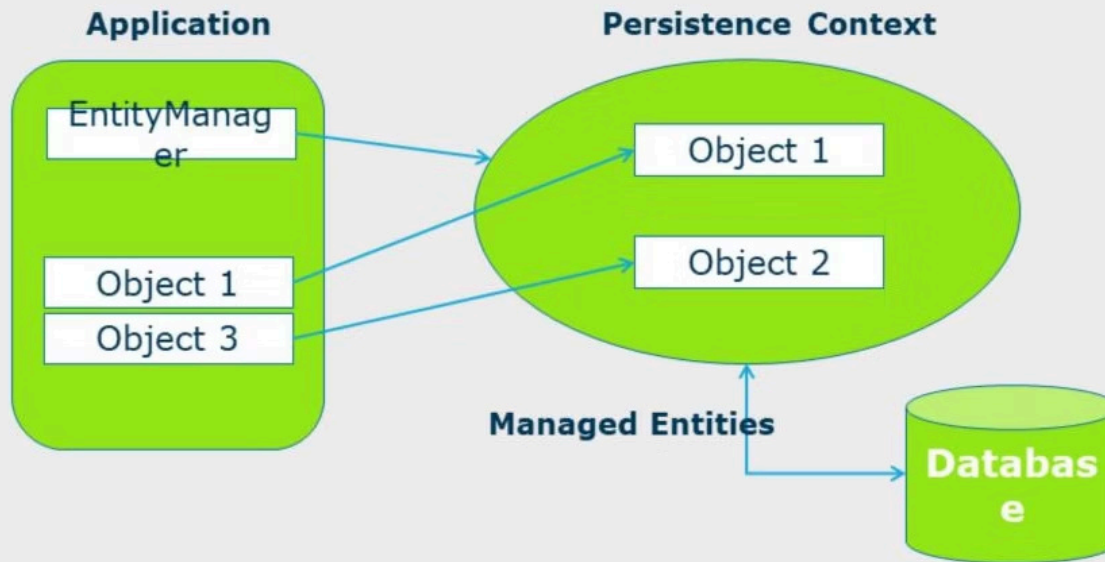
JPA

 by Praveen Kumar

JPA Runtime Overview

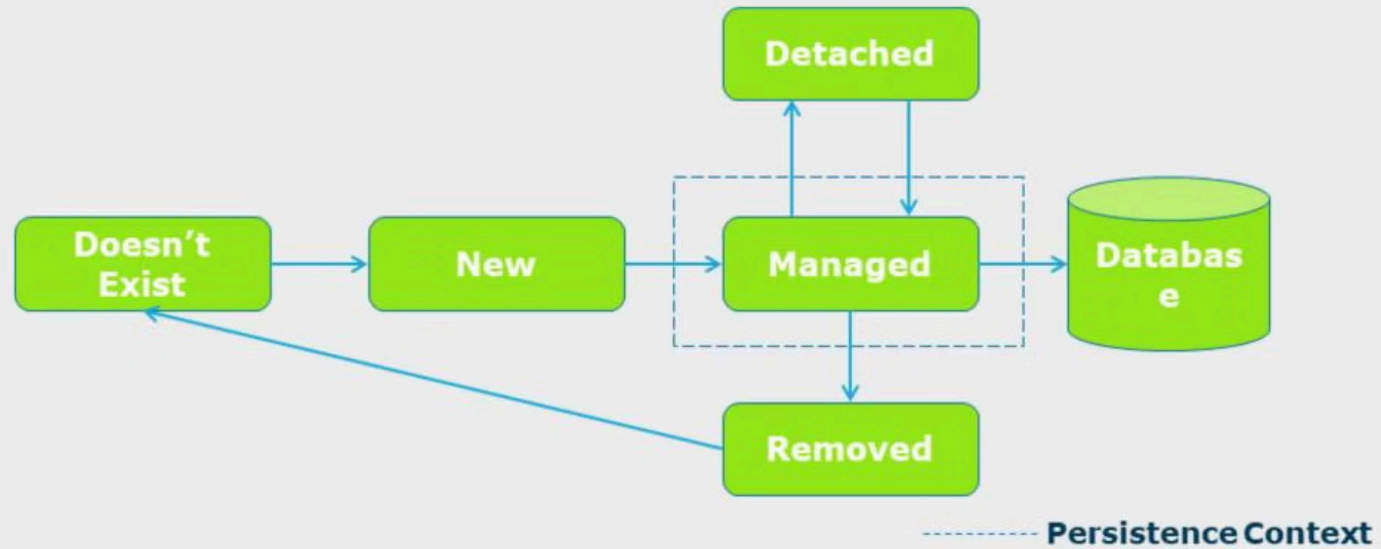
JPA Runtime Overview

Sample JPA Runtime:



Persistence Life Cycle

Object/Entity managed by ORM (using EntityManager) passes through different stages during its persistence.



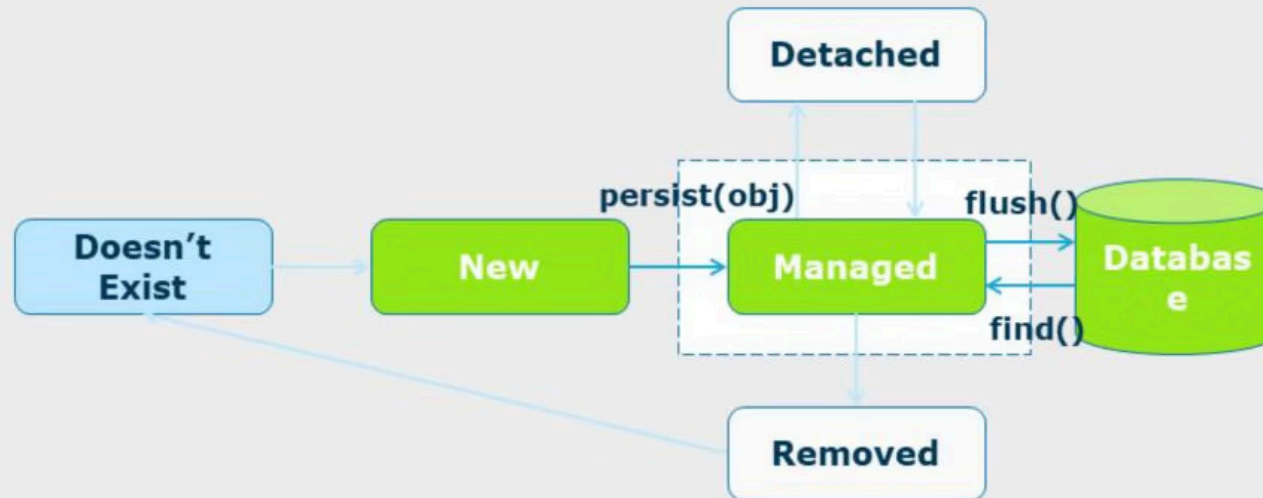
New / Transient State

An entity is new if it has just been instantiated using the new operator, and it is not associated with a persistence context. It has no persistent representation in the database and no identifier value has been assigned.



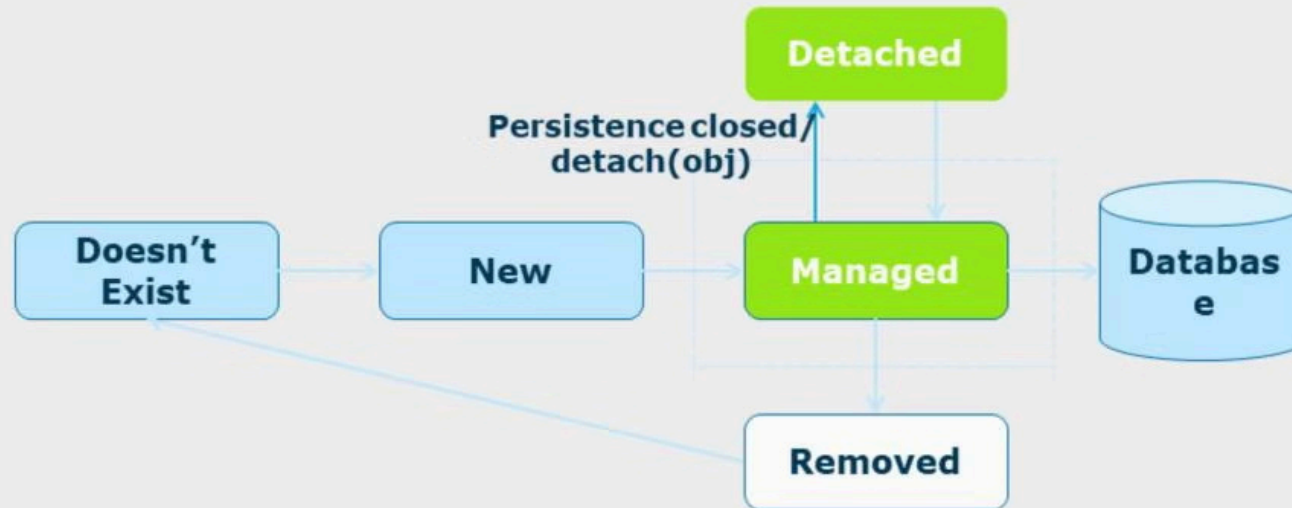
Managed State

A managed entity instance is an instance with a persistent identity that is currently associated with a persistence context.



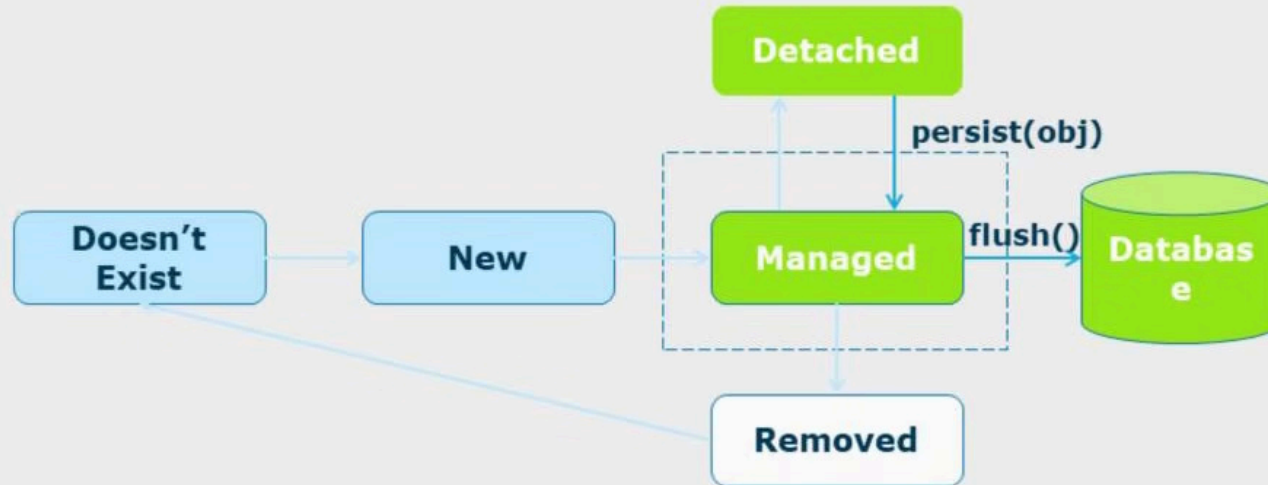
Detached State

The entity instance is an instance with a persistent identity that is no longer associated with a persistence context, usually because the persistence context was closed or the instance was evicted from the context.



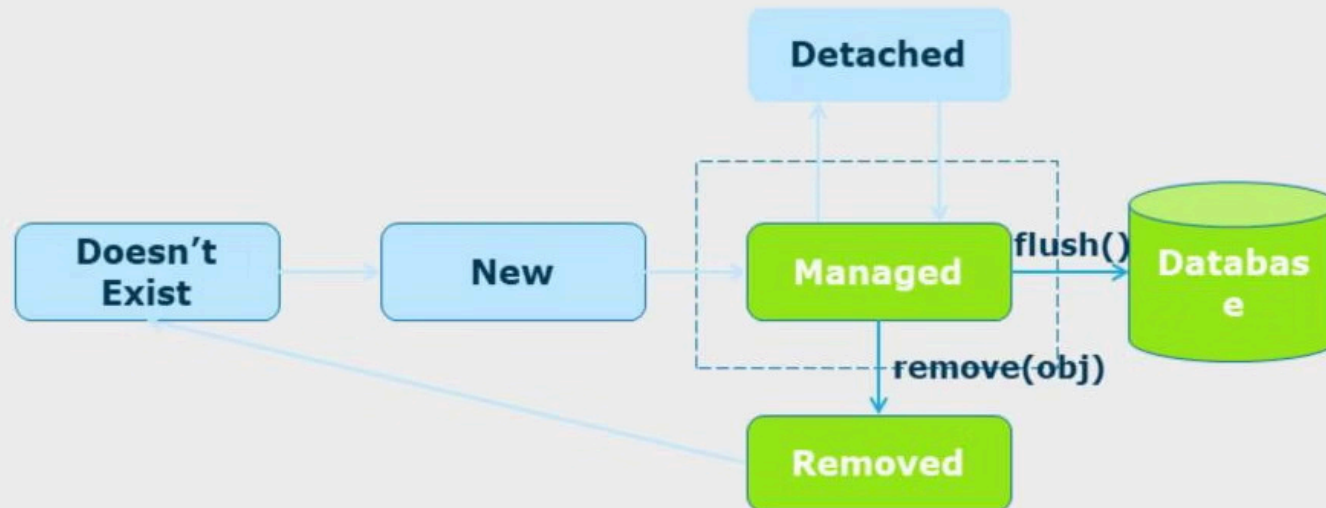
Detached State

A detached entity instance can be restored into managed state, when you persist it again. The ORM ensures state of entity instance is in sync with database.



Removed State

A removed entity instance is an instance with a persistent identity, associated with a persistence context, but scheduled for removal from the database



JPA Overview

Entity Classes

EntityManager

- Persistence Context

EntityManagerFactory

EntityTransaction

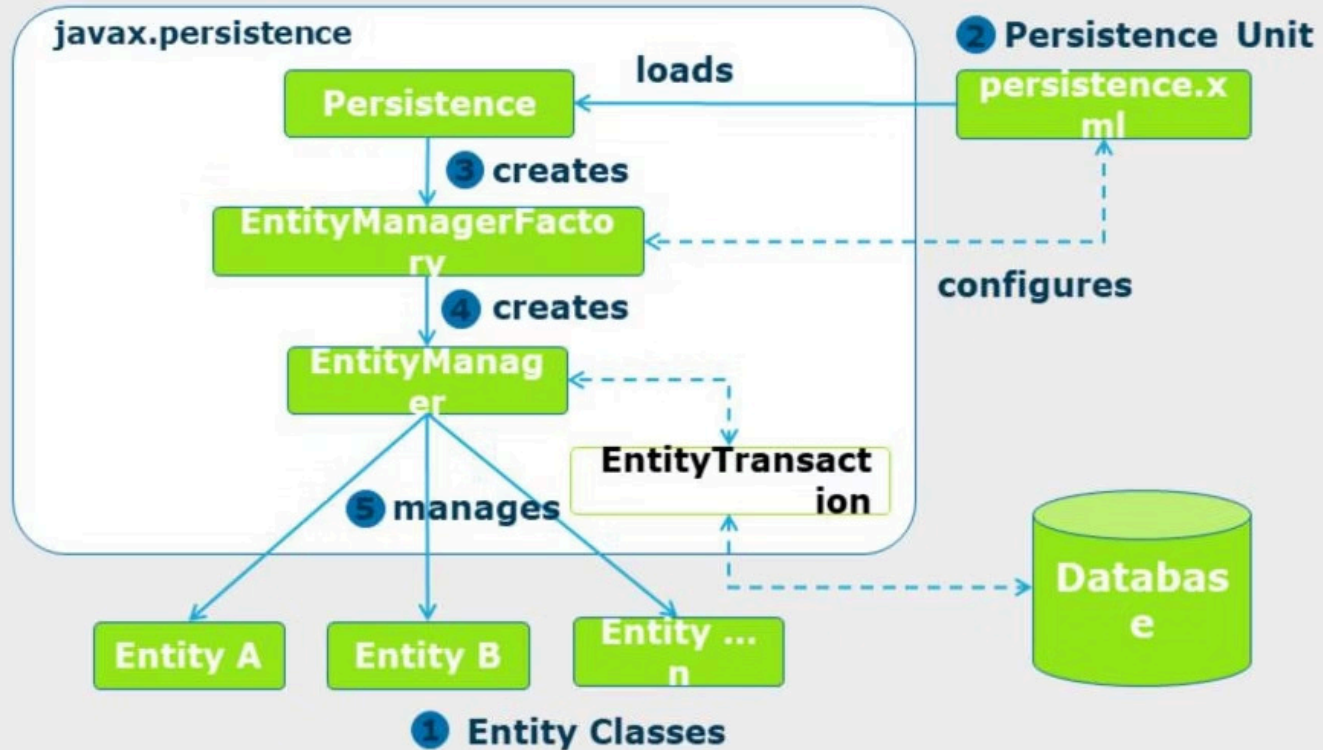
Persistence Unit

- persistence.xml

Java Persistence Query Language (JPQL)

- Query

Working with JPA



Creating Entity Classes

Java POJO classes can be made entity via either one of the following way:

- XML configuration (orm.xml)
- Annotations

A single xml file i.e. orm.xml is required to map entire set of entity classes within an application.

In case of Annotations, it should be marked in individual classes.

@Entity

```
public class Student implements Serializable {
```

@Id

```
private int studentId;
```

```
private String name;
```

```
//getters and setters
```

```
}
```

Entity Annotations

Mandatory Annotations

- @Entity
- @Id

Few more Annotations

- @GeneratedValue
- @Table
- @Column
- @Transient

@Entity

@Table(name="student_masters")

```
public class Student implements Serializable {
```

@Id

```
    @GeneratedValue(strategy=GenerationType.  
    AUTO)
```

```
    @Column(name = "stud_id")
```

```
    private int studentId;
```

```
    private String name;
```

Persistence Configuration

JPA persistence configuration is done with an XML file called "persistence.xml" which contains information about how to connect to the underlying database.

```
<persistence xmlns="http://java.sun.com/xml/ns/persistence"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://java.sun.com/xml/ns/persistence
http://java.sun.com/xml/ns/persistence/persistence\_2\_0.xsd
  version="2.0">
  <persistence-unit name="unit-name">
    <provider> <!-- JPA provider name like hibernate-->
  </provider>
    <properties> <!-- Database properties --> </properties>
  </persistence-unit>
</persistence>
```

Obtaining Entity Manager

An EntityManager is responsible for managing entities.

Below are the steps to create instance of EntityManager.

- Create EntityManagerFactory by using the Persistence class
- Call createEntityManager() on an EntityManagerFactory.

```
EntityManagerFactory emf =  
Persistence.createEntityManagerFactory(name);  
EntityManager em = emf.createEntityManager();
```

Working with Entity Manager

The EntityManager interface is providing the API for interacting with the Entity.

- Creates and removes persistent entity instances
- Finds entities by their primary key
- Allows for data querying
- Interacts with the persistence context

Following are some of the methods of EntityManager

Task	EntityManager method
Save new/detached entity	<code>persist(entity-instance)</code>
Update state of entity	<code>merge(entity-instance)</code>
Remove entity	<code>remove(entity-instance)</code>
Search for entity	<code>find(class,idvalue)</code>

Persisting entity with Entity Manager

```
public static void main(String[] args) {  
    EntityManagerFactory factory =  
        Persistence.createEntityManagerFactory("JPA-PU");  
    EntityManager em = factory.createEntityManager();  
    em.getTransaction().begin();  
    Student student = new Student();  
    student.setName("John");  
    em.persist(student);  
    em.getTransaction().commit();  
    em.close();  
    factory.close();  
}
```

```
@Entity  
public class Student implements  
    Serializable {  
    @Id  
    private int studentId;  
    private String name;  
}
```

Entity CRUD with Entity Manager

```
public Student getStudentById(int id) {  
    Student student = entityManager.find(Student.class, id);  
    return student;  
}  
  
public void addStudent(Student student) {  
    entityManager.persist(student);  
}  
  
public void removeStudent(Student student) {  
    entityManager.remove(student);  
}  
  
public void updateStudent(Student student) {  
    entityManager.merge(student);  
}
```

```
@Entity  
public class Student impl...  
{  
    @Id  
    private int studentId;  
    private String name;  
}
```